In 41–44 a set of premises and a conclusion are given. Use the valid argument forms listed in Table 2.3.1 to deduce the conclusion from the premises, giving a reason for each step as in Example 2.3.8. Assume all variables are statement variables.

42. **a.** $p \vee q$
b. $q \rightarrow r$
c. $p \wedge s \rightarrow t$
d. $\sim r$
e. $\sim q \rightarrow u \wedge s$
f. $\therefore t$

Let's first look at the valid argument forms listed in Table 2.3.1.

Modus Ponens	$p \rightarrow q$ p $\therefore q$	
Modus Tollens	$p \rightarrow q$ $\sim q$ $\therefore \sim p$	
Generalization	**a.** p $\therefore p \vee q$	**b.** q $\therefore p \vee q$
Specialization	**a.** $p \wedge q$ $\therefore p$	**b.** $p \wedge q$ $\therefore q$
Conjunction	p q $\therefore p \wedge q$	

CONT.:

Elimination	**a.** $p \vee q$ $\sim q$ $\therefore p$ **b.** $p \vee q$ $\sim p$ $\therefore q$
Transitivity	$p \to q$ $q \to r$ $\therefore p \to r$
Proof by Division into Cases	$p \vee q$ $p \to r$ $q \to r$ $\therefore r$
Contradiction rule	$\sim p \to \mathbf{c}$ $\therefore p$

Essentially, we start by assessing how to conclude t, and identify the closest condition for t, which happens to be in premise 3.

Here, we can establish that a prerequisite proof for t is p and s. So, the next natural step is to prove those statement variables.

From premise 1, we can prove p by rule of Elimination.

From premise 5, we can prove s by rule of Specialization. However, we need to prove ~q in order to properly prove s. This can be done by rule of Modus Tollens with premise 2 and premise 4.

See sequence below:

1. $p \vee q$
2. $q \rightarrow r$
3. $p \wedge s \rightarrow t$
4. $\sim r$
5. $\sim q \rightarrow u \wedge s$

$\therefore t$

6.	$\sim q$	Modus Tollens (2, 4)
7.	$u \wedge s$	Modus Ponens (5, 6)
8.	s	Specialization (7)
9.	p	Elimination (1, 6)
10.	$p \wedge s$	Conjunction (8, 9)
11.	t	Modus Ponens (3, 10)

$\therefore t$

Now that we've established a sequence of proofs, we can conclude t.

Solution: $\therefore t$